Random_forest Regression assignment

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SETTING A WORKING DIRECTORY

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(tidyverse)
library(rio)
library(ggplot2)
library(psych)
library(ggthemes)
library(summarytools)
library(tidyr)
library(formatR)
library(sjmisc)
library(caret)
library(report)
library(knitr)
library(randomForest)
library(reshape2)
```

Random forest regression: assignment quiz

import data set

```
df <- read.csv("rfregression.csv")
head(df)
```

```
  House_ID  Location Bedrooms Bathrooms House_Size_m2 Plot_Size_m2
1    H0001  Roysambu        4         3         290.3        532.9
2    H0002 Kitengela        6         5         344.6        505.0
3    H0003     Ruaka        6         4         420.0        647.5
4    H0004     Runda        3         3         256.6        389.6
5    H0005  Kilimani        3         2         131.3        264.1
6    H0006  Kilimani        5         4         361.9        601.5
  House_Age_Years Distance_CBD_km Parking_Spaces Has_Garden Condition
1            16.0            11.9              2        Yes Excellent
2             1.3            10.0              3        Yes      Good
3             6.6            11.7              1        Yes      Fair
4             6.0             6.1              1        Yes      Good
5            11.0             8.6              0         No      Good
6             3.5             8.7              1        Yes      Good
  House_Price_Million_KES
1                   74.60
2                   62.30
3                   60.17
4                   76.31
5                   41.35
6                   80.16
```

examine structure an dimension

```
str(df)
```

```
'data.frame':   800 obs. of  12 variables:
 $ House_ID              : chr  "H0001" "H0002" "H0003" "H0004" ...
 $ Location              : chr  "Roysambu" "Kitengela" "Ruaka" "Rund
a" ...
 $ Bedrooms              : int  4 6 6 3 3 5 4 3 5 2 ...
 $ Bathrooms             : int  3 5 4 3 2 4 4 2 4 2 ...
 $ House_Size_m2         : num  290 345 420 257 131 ...
 $ Plot_Size_m2          : num  533 505 648 390 264 ...
 $ House_Age_Years       : num  16 1.3 6.6 6 11 3.5 9.3 6.7 14.6 23.5
...
 $ Distance_CBD_km       : num  11.9 10 11.7 6.1 8.6 8.7 13.1 3.7 8.7
18.6 ...
 $ Parking_Spaces        : int  2 3 1 1 0 1 4 1 2 2 ...
 $ Has_Garden            : chr  "Yes" "Yes" "Yes" "Yes" ...
 $ Condition             : chr  "Excellent" "Good" "Fair" "Good" ...
 $ House_Price_Million_KES: num  74.6 62.3 60.2 76.3 41.4 ...
```

dimension

```
dim(df)
```

```
[1] 800  12
```

check for missing values

columwise and rowwise

```
sum(is.na(df))
```

```
[1] 0
```

```
cat("\n")
```

```
colSums(is.na(df))
```

```
                House_ID                 Location                 Bedroom
s
                       0                        0
0
               Bathrooms           House_Size_m2             Plot_Size_m
2
                       0                        0
0
        House_Age_Years         Distance_CBD_km          Parking_Space
s
                       0                        0
0
             Has_Garden               Condition House_Price_Million_KE
S
                       0                        0
0
```

Identify and handle duplicate observations

```
sum(duplicated(df))
```

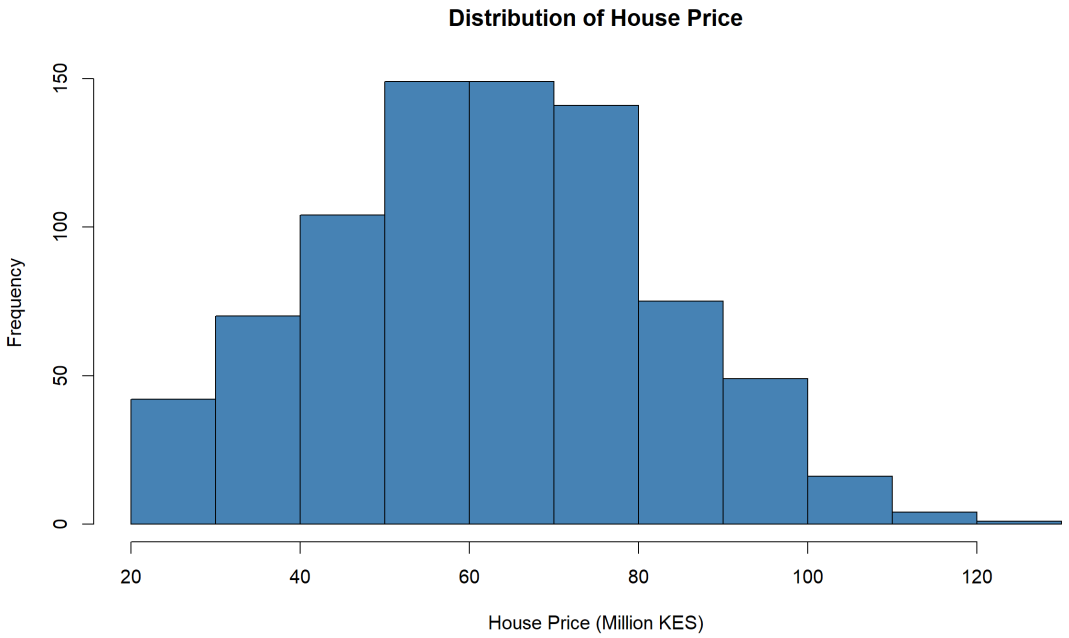
```
[1] 0
```

```
sum(duplicated(df[, !names(df) %in% "House_ID"]))
```

```
[1] 0
```

Explore distribution of target variable

```
hist(df$House_Price_Million_KES, main = "Distribution of House Price",
xlab = "House Price (Million KES)",
    col = "steelblue")
```




```
plot(density(df$House_Price_Million_KES), main = "Density of House Pri
ce", xlab = "House Price (Million KES)")
```



Examine relationships between predictors and house price ## Identify numeric variables and descriptive analysis

```
num_vars <- names(df)[sapply(df, is.numeric)]
desc_stats <- df %>%
    summarise(across(all_of(num_vars), list(n = ~sum(!is.na(.)), mean
= ~mean(.,
        na.rm = TRUE), sd = ~sd(., na.rm = TRUE), min = ~min(., na.rm
= TRUE), q25 = ~quantile(.,
        0.25, na.rm = TRUE), med = ~median(., na.rm = TRUE), q75 = ~qu
antile(., 0.75,
        na.rm = TRUE), max = ~max(., na.rm = TRUE)))) %>%
    pivot_longer(everything(), names_to = c("variable", "stat"), names
_sep = "_(?=[^_]+$)") %>%
    pivot_wider(names_from = stat, values_from = value) %>%
    arrange(variable)
print(desc_stats)
```

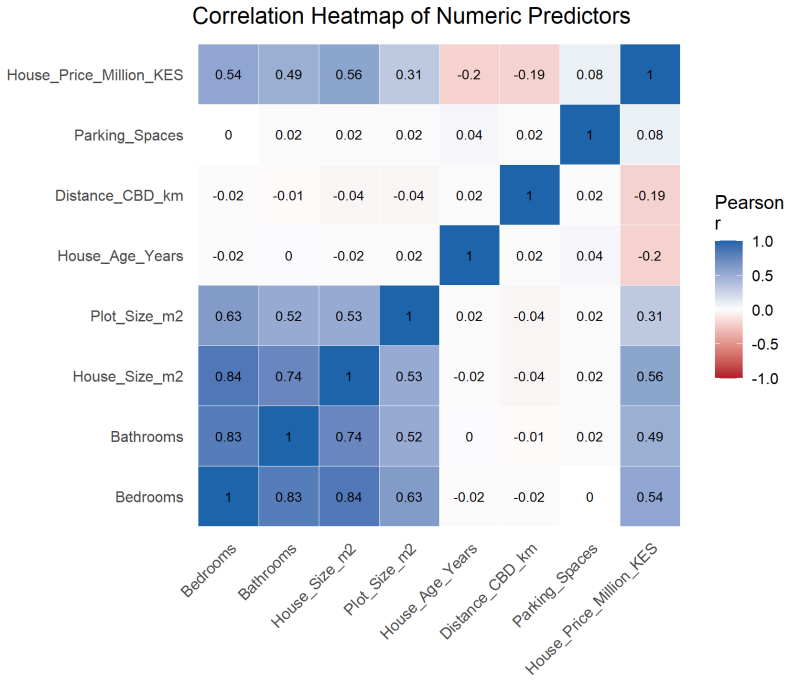
```
# A tibble: 8 × 9
  variable                    n   mean     sd   min    q25   med   q75
max
  <chr>                   <dbl>  <dbl>  <dbl> <dbl>  <dbl> <dbl> <dbl>
<dbl>
1 Bathrooms                 800   3.13   1.15     1   2      3     4
5
2 Bedrooms                  800   3.88   1.39     2   3      4     5
6
3 Distance_CBD_km           800   9.17   4.91     1   5.4    9    12.6
24.6
4 House_Age_Years           800  12.0    7.65     1   6.38  10.2  16.5
35
5 House_Price_Million_KES   800  62.5   19.5     25  48.8   62.5  75.9
121.
6 House_Size_m2             800 265.    81.4     70 204.   263.  326.
420
7 Parking_Spaces            800   1.40   1.13     0   1      1     2
4
8 Plot_Size_m2              800 464.   157.     180 353.   456.  570.
967.
```

Correlation among numeric predictors

```
cor_num <- cor(df[, num_vars], use = "complete.obs")
print(round(cor_num, 2))
```

```
                        Bedrooms Bathrooms House_Size_m2 Plot_Size_m2
Bedrooms                    1.00      0.83          0.84         0.63
Bathrooms                   0.83      1.00          0.74         0.52
House_Size_m2               0.84      0.74          1.00         0.53
Plot_Size_m2                0.63      0.52          0.53         1.00
House_Age_Years            -0.02      0.00         -0.02         0.02
Distance_CBD_km            -0.02     -0.01         -0.04        -0.04
Parking_Spaces              0.00      0.02          0.02         0.02
House_Price_Million_KES     0.54      0.49          0.56         0.31
                        House_Age_Years Distance_CBD_km Parking_Spaces
Bedrooms                          -0.02           -0.02           0.00
Bathrooms                          0.00           -0.01           0.02
House_Size_m2                     -0.02           -0.04           0.02
Plot_Size_m2                       0.02           -0.04           0.02
House_Age_Years                    1.00            0.02           0.04
Distance_CBD_km                    0.02            1.00           0.02
Parking_Spaces                     0.04            0.02           1.00
House_Price_Million_KES           -0.20           -0.19           0.08
                        House_Price_Million_KES
Bedrooms                                   0.54
Bathrooms                                  0.49
House_Size_m2                              0.56
Plot_Size_m2                               0.31
House_Age_Years                           -0.20
Distance_CBD_km                           -0.19
Parking_Spaces                             0.08
House_Price_Million_KES                    1.00
```

```
cor_target <- cor(df[, num_vars], df$House_Price_Million_KES, use = "c
omplete.obs")
cor_melt <- melt(cor_num)
colnames(cor_melt) <- c("Var1", "Var2", "Correlation")
ggplot(cor_melt, aes(x = Var1, y = Var2, fill = Correlation)) + geom_t
ile(color = "white") +
    geom_text(aes(label = round(Correlation, 2)), size = 3, color = "b
lack") + scale_fill_gradient2(low = "#B2182B",
    mid = "white", high = "#2166AC", midpoint = 0, limits = c(-1, 1),
name = "Pearson\nr") +
    labs(title = "Correlation Heatmap of Numeric Predictors", x = NUL
L, y = NULL) +
    theme_minimal(base_size = 12) + theme(axis.text.x = element_text(a
ngle = 45,
    hjust = 1), panel.grid = element_blank()) + coord_fixed()
```



Convert categorical variables into numerical representations

```
df_model <- df %>%
    select(-House_ID)
dummies <- dummyVars(House_Price_Million_KES ~ ., data = df_model)
df_encoded <- predict(dummies, newdata = df_model) %>%
    as.data.frame()
df_encoded$House_Price_Million_KES <- df_model$House_Price_Million_KES
str(df_encoded)
```

```
'data.frame':   800 obs. of  21 variables:
 $ LocationKasarani       : num  0 0 0 0 0 0 0 0 0 0 ...
 $ LocationKileleshwa     : num  0 0 0 0 0 0 1 0 0 0 ...
 $ LocationKilimani       : num  0 0 0 0 1 1 0 0 0 0 ...
 $ LocationKitengela      : num  0 1 0 0 0 0 0 1 0 0 ...
 $ LocationRoysambu       : num  1 0 0 0 0 0 0 0 0 0 ...
 $ LocationRuaka          : num  0 0 1 0 0 0 0 0 0 0 ...
 $ LocationRunda          : num  0 0 0 1 0 0 0 0 0 0 ...
 $ LocationSyokimau       : num  0 0 0 0 0 0 0 0 1 1 ...
 $ Bedrooms               : num  4 6 6 3 3 5 4 3 5 2 ...
 $ Bathrooms              : num  3 5 4 3 2 4 4 2 4 2 ...
 $ House_Size_m2          : num  290 345 420 257 131 ...
 $ Plot_Size_m2           : num  533 505 648 390 264 ...
 $ House_Age_Years        : num  16 1.3 6.6 6 11 3.5 9.3 6.7 14.6 23.5
...
 $ Distance_CBD_km        : num  11.9 10 11.7 6.1 8.6 8.7 13.1 3.7 8.7
18.6 ...
 $ Parking_Spaces         : num  2 3 1 1 0 1 4 1 2 2 ...
 $ Has_GardenNo           : num  0 0 0 0 1 0 0 0 0 0 ...
 $ Has_GardenYes          : num  1 1 1 1 0 1 1 1 1 1 ...
 $ ConditionExcellent     : num  1 0 0 0 0 0 1 0 1 0 ...
 $ ConditionFair          : num  0 0 1 0 0 0 0 0 0 1 ...
 $ ConditionGood          : num  0 1 0 1 1 1 0 1 0 0 ...
 $ House_Price_Million_KES: num  74.6 62.3 60.2 76.3 41.4 ...
```

Divide data into training and testing sets

```
set.seed(123)
train_index <- createDataPartition(df_encoded$House_Price_Million_KES,
p = 0.8, list = FALSE)
train <- df_encoded[train_index, ]
test <- df_encoded[-train_index, ]
cat("Train rows:", nrow(train), " Test rows:", nrow(test), "\n")
```

```
Train rows: 640  Test rows: 160
```

Build Random Forest Regression model

```
set.seed(123)
rf_model <- randomForest(House_Price_Million_KES ~ ., data = train, nt
ree = 500,
    mtry = 3, importance = TRUE)
print(rf_model)
```

```

Call:
 randomForest(formula = House_Price_Million_KES ~ ., data = train,
ntree = 500, mtry = 3, importance = TRUE)
               Type of random forest: regression
                     Number of trees: 500
No. of variables tried at each split: 3

          Mean of squared residuals: 115.021
                    % Var explained: 69.19
```

predictions for test data

```
pred <- predict(rf_model, newdata = test)
head(pred)
```

```
       1        2        8       19       20       31
70.17076 69.67807 50.20328 67.86424 49.59666 58.61508
```

Evaluate model: RMSE, MAE, R2, MAPE

```
actual <- test$House_Price_Million_KES
rmse <- sqrt(mean((actual - pred)^2))
mae <- mean(abs(actual - pred))
r2 <- 1 - sum((actual - pred)^2)/sum((actual - mean(actual))^2)
mape <- mean(abs((actual - pred)/actual)) * 100
metrics <- c(RMSE = rmse, MAE = mae, R2 = r2, MAPE = mape)
print(round(metrics, 3))
```

```
  RMSE    MAE     R2   MAPE
10.726  8.423  0.718 15.016
```

Compare actual and predicted house prices

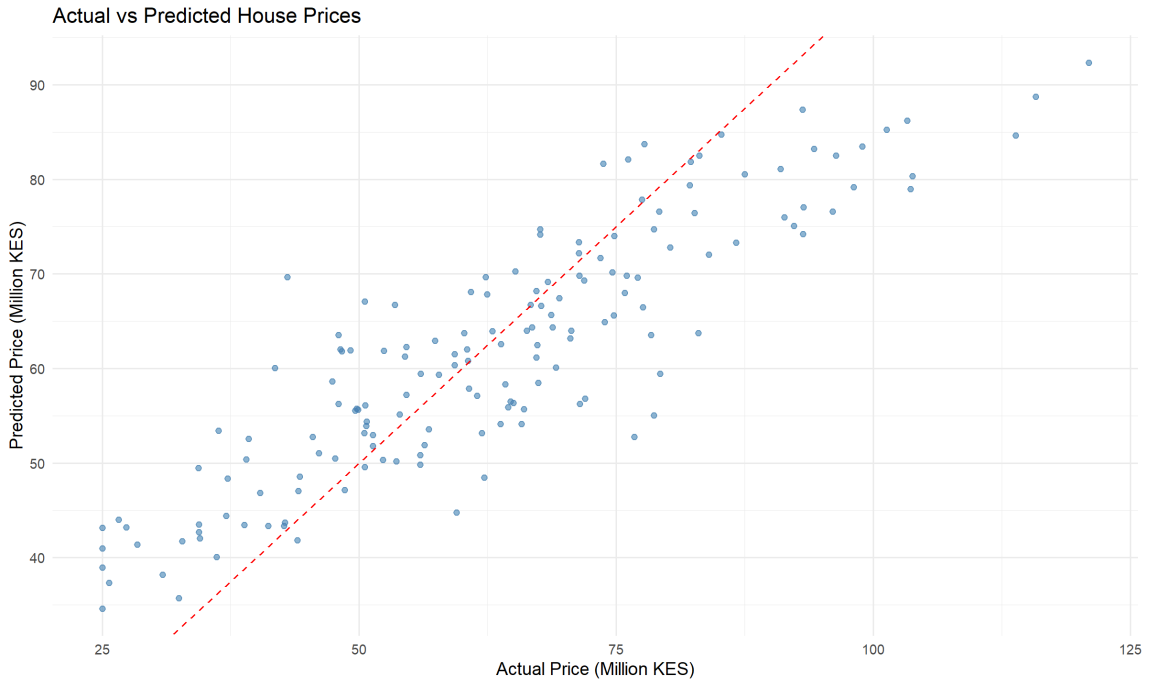
```
comparison <- data.frame(Actual = actual, Predicted = pred, Residual =
actual - pred)
head(comparison)
```

```
   Actual Predicted    Residual
1   74.60  70.17076   4.4292437
2   62.30  69.67807  -7.3780706
8   53.63  50.20328   3.4267174
19  62.45  67.86424  -5.4142370
20  50.56  49.59666   0.9633406
31  47.37  58.61508 -11.2450812
```

```
write.csv(comparison, "actual_vs_predicted.csv", row.names = FALSE)
```

Plot Actual vs Predicted values

```
ggplot(comparison, aes(x = Actual, y = Predicted)) + geom_point(alpha
= 0.6, color = "steelblue") +
    geom_abline(intercept = 0, slope = 1, color = "red", linetype = "d
ashed") + labs(title = "Actual vs Predicted House Prices",
    x = "Actual Price (Million KES)", y = "Predicted Price (Million KE
S)") + theme_minimal()
```

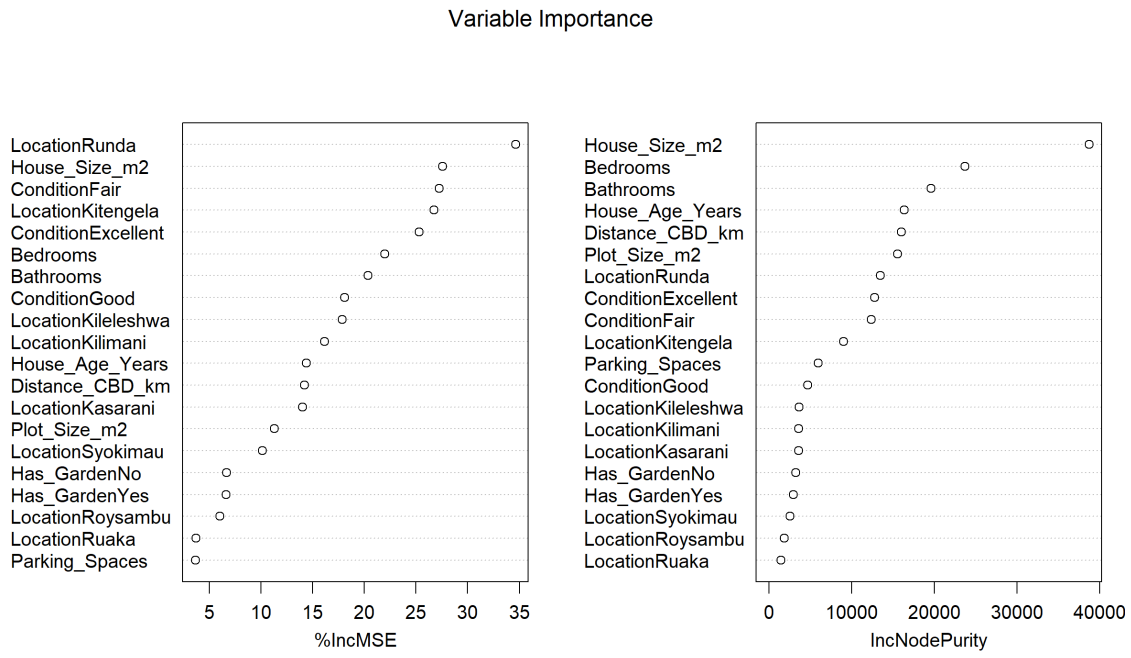


Examine variable importance

```
imp <- importance(rf_model)
print(imp[order(imp[, "%IncMSE"], decreasing = TRUE), ])
```

```
                     %IncMSE IncNodePurity
LocationRunda      34.640143     13436.736
House_Size_m2      27.588855     38728.844
ConditionFair      27.265507     12368.075
LocationKitengela  26.760288      9015.462
ConditionExcellent 25.322603     12792.628
Bedrooms           21.977215     23700.317
Bathrooms          20.348734     19586.704
ConditionGood      18.110325      4679.970
LocationKileleshwa 17.866907      3653.206
LocationKilimani   16.166703      3585.251
House_Age_Years    14.377249     16348.246
Distance_CBD_km    14.215435     16034.148
LocationKasarani   14.009328      3553.790
Plot_Size_m2       11.281674     15546.698
LocationSyokimau   10.121013      2510.172
Has_GardenNo        6.662746      3216.090
Has_GardenYes       6.614218      2942.833
LocationRoysambu    6.034947      1834.734
LocationRuaka       3.734109      1438.152
Parking_Spaces      3.682755      5959.029
```

```
varImpPlot(rf_model, main = "Variable Importance")
```



18. The most important predictors

- The most important predictors are LocationRunda, House_Size_m2, and ConditionFair, which lead on both %IncMSE (34.6, 27.6, 27.3) and IncNodePurity (13,437; 38,729; 12,368), indicating they drive both predictive accuracy and node splits. They are closely followed by LocationKitengela, ConditionExcellent, Bedrooms, and Bathrooms (all with %IncMSE above 20), showing that location, size, condition, and room count dominate house price variation. Mid-tier predictors include ConditionGood, LocationKileleshwa, LocationKilimani, House_Age_Years, and Distance_CBD_km (%IncMSE ~14–18), while Plot_Size_m2 and LocationSyokimau contribute modestly. The least important predictors are Has_Garden (both levels), LocationRoysambu, LocationRuaka, and Parking_Spaces (%IncMSE < 7), suggesting they add little unique information once location, size, and condition are accounted for.